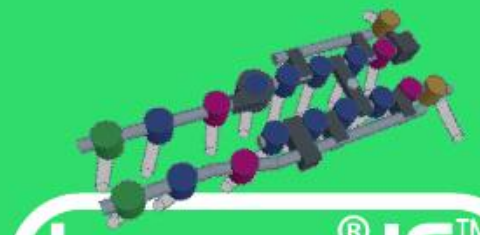
CRENTEC
IMPLANT SPECIALIST



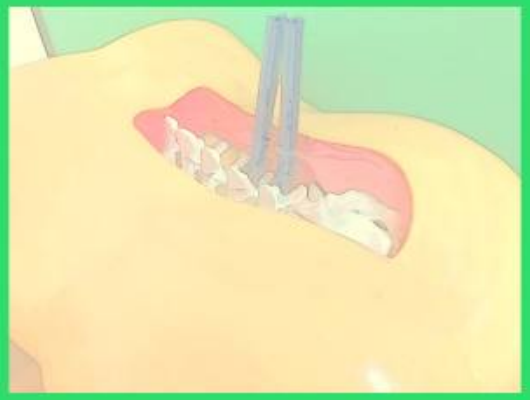
Lospa® IS™

Rhino™

Percutaneous Pedicle Screw System
Surgical Technique



The LOSPA® IS™ Rhino™ Percutaneous Pedicle Screw System is a simple solution for posterior thoracolumbar surgical cases especially, degenerative disc disease procedures. This exceptional system evolved from listening to the demanding requirements and insightful suggestions of industry leading spine surgeons. Corentec Co., Ltd. would like to take this opportunity to thank them for their contributions. Their efforts helped us deliver a versatile selection of devices and ergonomically-designed instrument sets.



Introduction

Features

Patient Position

Percutaneous Pedicle Trajectory

Pedicle Marking

Jamshidi Needle Placement

K-wire Placement

S1 Pedicle Targeting

Dilation and Taping

Screw Driver Assembly

Screw Insertion

Screw Adjusting and Rod Preparation

Rod Insertion and Reduction

Compression

Final Tightening

Postoperative Care

Product Information

Rhino™

Percutaneous Pedicle Screw System

Surgical Technique

Introduction

Description

The Rhino™ Percutaneous Pedicle Screw System is a temporary, multiple component system comprised of a variety of non-sterile, single use components, made of titanium alloy, that allow the surgeon to build a spinal implant construct minimizing muscle injury. The system is attached to the vertebral body by means of screw fixation to the non-cervical spine. The Rhino™ Percutaneous Pedicle Screw System consists of an assortment of cannulated long tab pedicle screws, set screws. The Percutaneous Pedicle Screw System implants are not compatible with components or metal from any other manufacturer's system.

Indications

Rhino™ Percutaneous Pedicle Screw System is intended for posterior, non-cervical pedicle fixation. The system is limited to skeletally mature patients and is intended to be used as an adjunct to fusion using autograft or allograft. The device is indicated for all of the following indications: 1. degenerative disc disease (defined as discogenic back pain with degeneration of the disc confirmed by history and radiographic studies) 2. spondylolisthesis, 3. trauma (i.e., fracture or dislocation), 4. spinal stenosis, 5. deformities or curvatures (i.e., scoliosis, kyphosis, and/or lordosis), 6. tumor, 7. pseudoarthrosis, and 8. failed previous fusion

Contraindications include, but are not limited to

1. Morbid obesity 2. Mental Illness 3. Alcoholism or drug abuse 4. Pregnancy 5. Metal sensitivity/allergies 6. Severe osteopenia 7. Patients unwilling or unable to follow post-operative care instructions 8. Any circumstances not listed under the heading indications.

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Percutaneous Pedicle Screw System

Surgical Technique

Potential Adverse Events

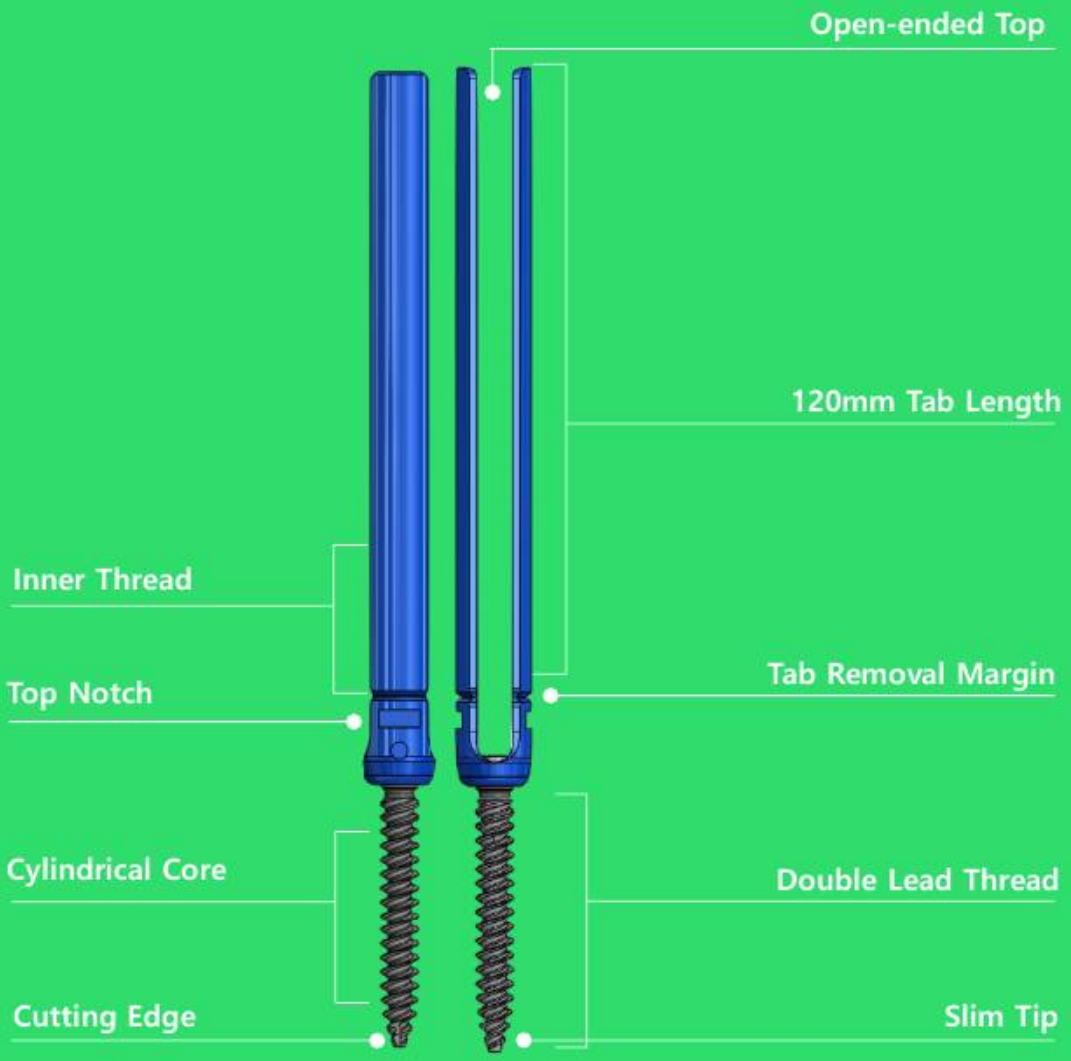
All of the possible adverse events associated with spinal fusion surgery without instrumentation are possible. With instrumentation, a listing of possible adverse events includes, but is not limited to: 1. Device component fracture 2. Loss of fixation 3. Non-union 4. Fracture of the vertebra 5. Neurological injury 6. Vascular or visceral injury 7. Early or late loosening of any or all of the components 8. Disassembly and/or bending of any or all components 9. Foreign body (allergic) reaction to implants, debris, corrosion products, graft material, including metallosis, straining, tumor formation, and/or auto-immune disease 10. Pressure on the skin from component parts in patients with inadequate tissue coverage over the implant possibly causing skin penetration, irritation, and/or pain 11. Post-operative change in spinal curvature, loss of correction, height, and/or reduction 12. Infection 13. Pain, discomfort, or abnormal sensations due to the presence of the device 14. Hemorrhage 15. Cessation of any potential growth of the operated portion of the spine 16. Death 17. Potential risks identified with the use of the device system may require additional surgery. 18. Excessive torque applied to the screws may strip the threads in the bone. 19. DO NOT REUSE IMPLANTS. Discard used, damaged, or otherwise suspect implants. 20. The implantation of pedicle screw spinal systems should be performed only by experienced spinal surgeons with specific training in the use of this pedicle screw spinal system because this is a technically demanding procedure presenting a risk of serious injury to the patient. 21. Based on fatigue testing results, the physician/surgeon should consider the levels of implantation, patient weight, patient activity level, other patient conditions, etc. which may impact on the performance of the system. 22. Mixing of dissimilar metals can accelerate the corrosion process. Do not use the titanium alloy or cobalt chrome alloy components of this system with implants of other material composition or components from different manufacturers unless specifically stated.

Rhino™

Percutaneous Pedicle Screw System
Surgical Technique

Features

1. Screw Head Profile
   (Diameter 13.3mm, Height 15.5mm)
2. Screw Thread Pitch(interval) 2mm
3. Polyaxial Screw with 50° cone angulation
4. Cannulated Screw, Ø1.8mm
5. Accommodating both 5.5 and 6.0 Rod
6. Long Tab Screw with extended Inner Thread
7. Easy Entry-Inserting via Self-Tap and Slim Tip
8. Double Lead Screw Thread for screwing in fast
9. Cylindrical Core with better pull-out strength
10. Implants and Instrument are fully combined with Lospa® IS™ Open Pedicle Screw System for Revision and Hybrid Surgery



Rhino™
Percutaneous Pedicle Screw System
Surgical Technique

Patient Position

The patient is placed on the operating room table in a prone position. If you place the patient on a frame with the hips extended, lumbar lordosis will increase. Radiographic imaging or quality X-rays are used intraoperatively. The patient's position should be checked radiographically (C-arm or X-ray) to determine the direction of the pedicle relative to the horizontal plane.



Rhino™

Percutaneous Pedicle Screw System

Surgical Technique

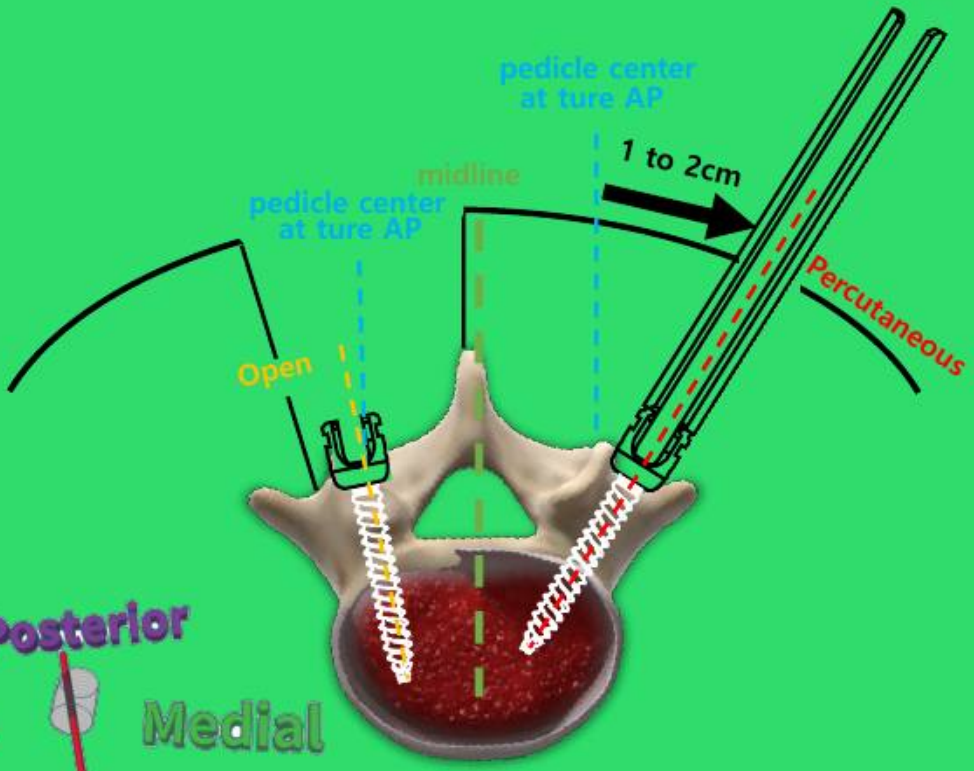
Percutaneous Pedicle Trajectory

Consider the pedicle as roughly a cylindrical structure. As the pedicle is traversed, the trajectory should allow the needle or screw to remain lateral to the medial pedicle wall. The ideal starting point is at the intersection of the facet and the transverse process (the lateral edge of the cylinder).

Spinal needle may be used to verify the appropriate location of the incisions. The needle is positioned on the paraspinal fascia directly over the pedicle on an AP image. The needle is then moved laterally 1cm to 2cm and inserted to the intersection of the facet and transverse process. The fascial incision is slightly lateral to the pedicle on fluoroscopy. This will help to ensure the needle follows the normal lateral to medial trajectory of the pedicle. Both AP and lateral images confirm that the appropriate starting place has been determined





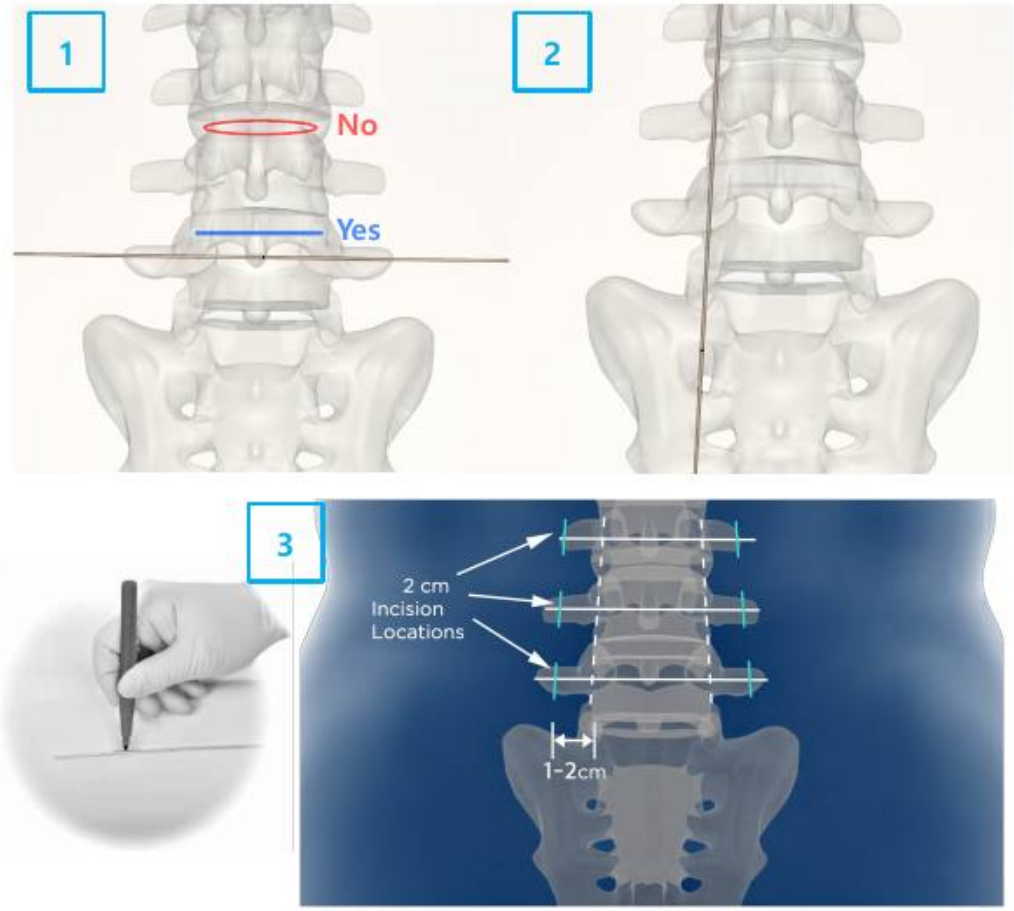


Pedicle Marking

Adjust orientation of the C-Arm until a true A/P view is obtained (endplates parallel, spinous process centered between pedicle centers) Place a guidewire across the vertebral body so that its projection splits the pedicle in the A/P fluoro image. Mark these locations on the skin. On AP and lateral fluoroscopy, the end plates should be linear and not rounded.

Place a guidewire oriented in the cephalad-caudal direction so that its projection is in-line with the lateral pedicle wall of the targeted level and adjacent levels. This plane allows for the pedicle screws to lay in-line for easy rod placement. Mark this location on the skin.

Make a second mark 1-2 cm lateral to the targeted pedicle location. This is the incision location. Repeat for the remaining pedicles to be targeted.



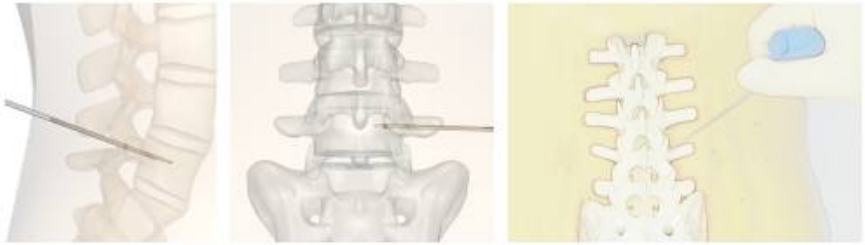
Jamshidi Needle Placement

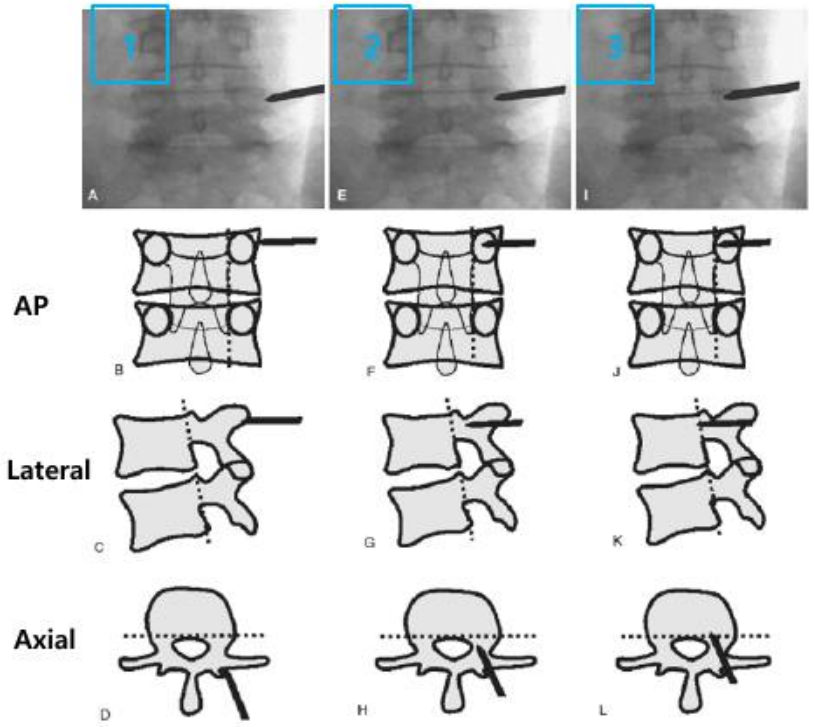
Make a 2.0cm longitudinal incision at location marked on the patient. Insert Jamshidi needle to contact the pedicle surface. ***Pulling back the trochar from the Jamshidi can allow the sheath-tip to be used to palpate the bony anatomy.***

Confirm position and orientation of Jamshidi by using fluoroscopy. Entry should be centered along the lateral border of the pedicle in the A/P view.

Using A/P fluoroscopy, advance the Jamshidi tip toward the pedicle's center. The tip of the Jamshidi should not extend beyond three-quarters across the pedicle.

Using lateral fluoroscopy, confirm the Jamshidi needle tip enters the vertebral body and remains centered in the pedicle. Advance the Jamshidi until its tip is within the vertebral body.



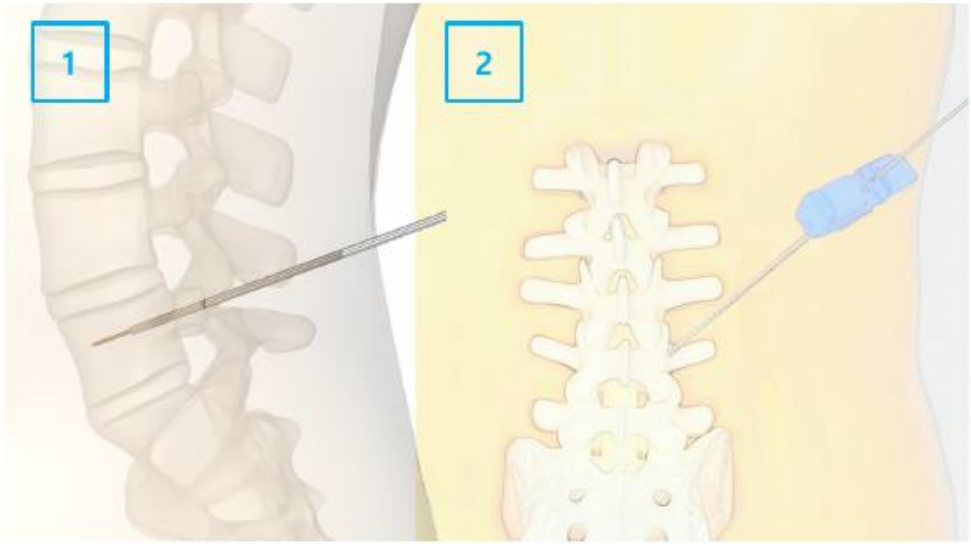


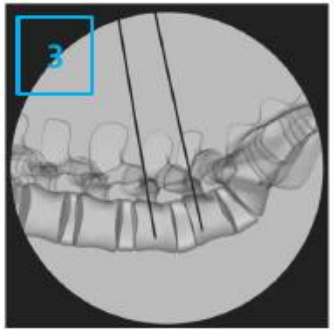
K-wire Placement

Remove inner stylet from the Jamshidi. Place K-wire through the Jamshidi and advance it into the vertebral body at or 2-3mm beyond the desired pedicle screw tip location. The provided K-wire driver can be used to assist in pushing the K-wire into place.

Visualize the K-wire with x-ray. The wire should be straight, without any bend or curve. A curve could cause cannulated instruments or the pedicle screw to bind on the wire and inadvertently advance the K-wire anteriorly, beyond the anterior cortex of the vertebral body into the patient's abdomen.

Carefully remove the JamShidi while holding the K-Wire to ensure it remains in position. Repeat K-wire placement on contralateral side.



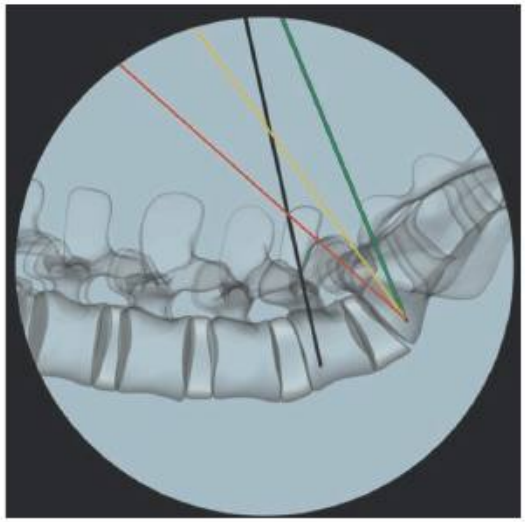


S1 Pedicle Targeting

When operating on S1, position the K-wires according to the green-colored K-wire (see image). Most surgeons will place S1 screws bicortical (i.e. just through the anterior cortex of S1).

Some tables have pedestals that make it difficult to get a true AP view of the pedicles, especially at the S1 level. While adjustments in patient positioning can be made, tables that limit good AP fluoroscopy should generally be avoided. A longer prep area is also necessary for intraoperative flexibility. Either a radiolucent frame or chest rolls may be used, but a knee-to-chest position should be avoided. Verify that adequate fluoroscopic images of the pedicles can be obtained in both an AP and lateral view before proceeding. To adequately view the S1 pedicle, a Ferguson view presents the best image.

The starting point in the sacral pedicles is different from the lumbar pedicles due to the lack of transverse processes and the presence of the sacral ala. The size and configuration of the S1 pedicle allow the surgeon more flexibility in positioning the screw within the sacrum. The S1 pedicle is caudal and slightly lateral to the superior articular process; therefore, the entry point should be in the most caudal portion of the pedicle.



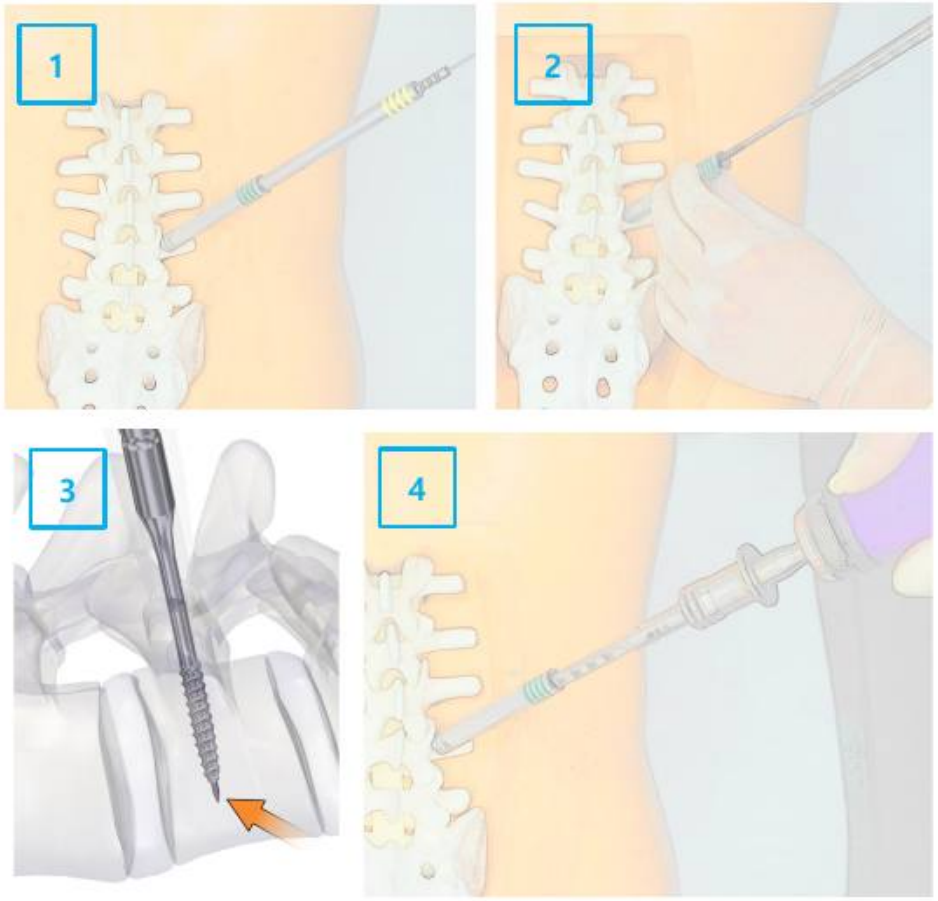


Dilation and Taping

Insert Dilator #1 over the K-Wire until the tip contacts the pedicle. Slide Dilator #2 & 3 over Dilator #1 until the tip contacts the pedicle. Remove Dilator #1 & 2 while holding the K-wire in place. Dilator #3 is made of a radiolucent polymer for ease of instrument visualization and nerve monitoring.

The fascia and muscle must be dilated to allow for screw placement. Incise the fascia to allow for dilation of the large dilator. Three dilators are used to gently make a path of the appropriate dimension. Dilators should dock on bony anatomy to minimize tissue creepage. The first two dilators are removed, leaving the third dilator to serve as a tissue protection sleeve during the tapping step.

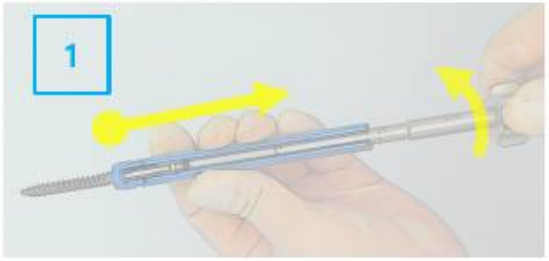
Tap depth is measured from the top of the #3 dilator to the tap's depth marks. Fluoroscopy should be used to visualize the tap and K-wire positions during tapping. Use care to ensure the wire is not bent, kinked or advanced anterior during tapping. Do not advance the tap beyond the tip of the K-wire to prevent bone from plugging the cannula, rendering the tap non-functional for subsequent pedicles. Minimize toggling of the tap to prevent the tap from bending or breaking.

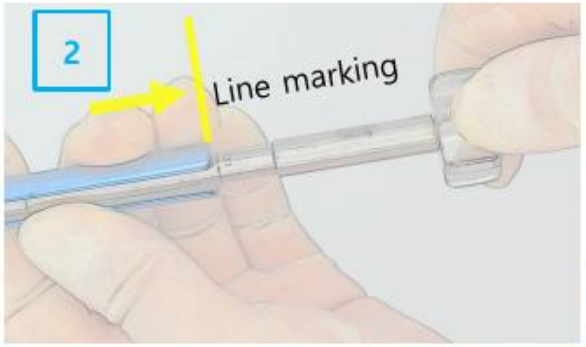


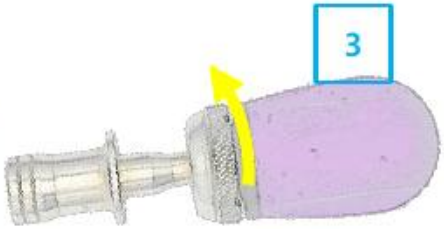
Screw Driver Assembly

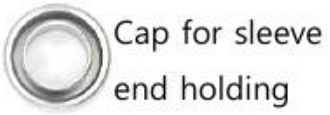
Select appropriate screw driver with respect to the pedicle screw being implanted. Insert the distal tip of driver into the body of the pedicle screw. Turn the knob clockwise to thread and secure the pedicle screw to the screw driver tip. Make sure the laser marking line 'Long' to meet the screw sleeve end. Ratchet mode should be forward.

Ensure the pedicle screw is rigidly fixed on the screw driver tip and is in alignment with the driver shaft prior to screw insertion. Misalignment, improper engagement of screw to driver, or loosening of knob during screw insertion can result in undesired trajectory of pedicle screw. Do not apply levering force to driver during screw insertion as this can result in missed trajectory of screw or pedicle fractures.









Cap for sleeve end holding

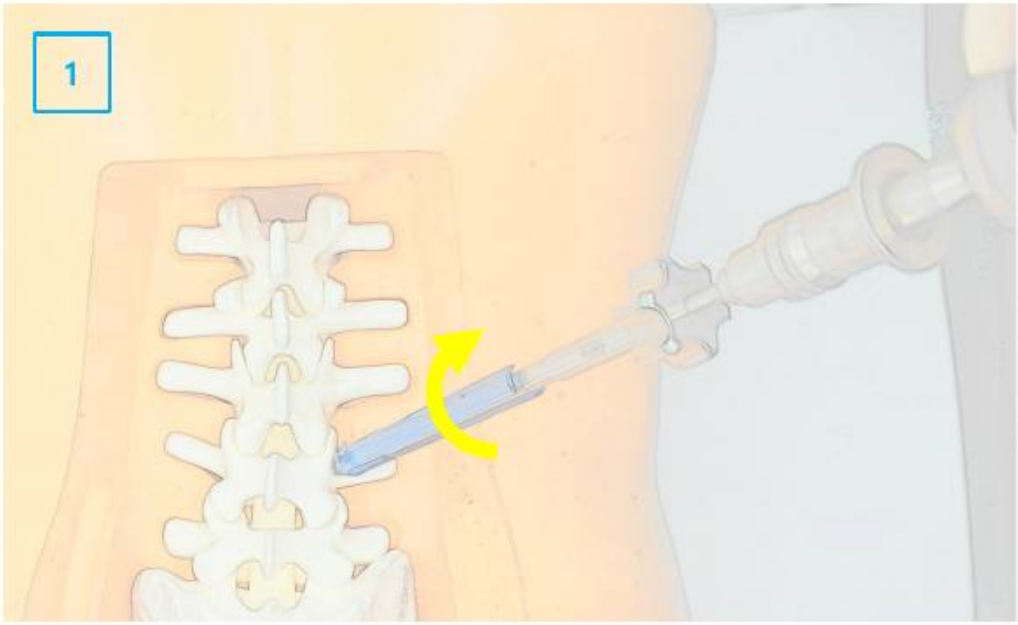


Screw Insertion

Advance the assembly over the K-wire until the screw contacts the pedicle surface, then advance the screw until the head lightly contacts the pedicle.

Once the screw tip has been advanced into the pedicle, the K-wire should be removed to ensure the wire is not advanced anterior. Binding the screw head against the pedicle will eliminate polyaxial motion of the screw and increase the difficulty of rod placement.

Unlock the thumbwheel, then rotate the thumbwheel counterclockwise until the pedicle driver can be removed. Maintain colinear alignment with the screw to ease driver removal.







Screw Adjusting and Rod Preparation

Pedicle Screw Adjustor can be used to adjust the height of the screw after the Pedicle Screw Driver has been removed. The Head Positioning is to adjust orientation of the extended tabs.

Extend the skin incision 1cm to rod entrance. Measuring rod length by using rod template. Lay the Rod Template on the skin next to the extended tab. Bend the Template accordingly to ensure all portions of the rod measurement device are in contact with the skin. Read the measurement off the Template to estimate the length of the rod to be inserted.

Pre-bent rods are provided with an alignment line along the inside of the curve. After choosing the appropriate length rod, contour and corresponding rod holder, insert the connection end of the rod into the pocket of the rod holder ensuring that the notch on the connection end of the rod is facing up towards the handle of the rod holder.

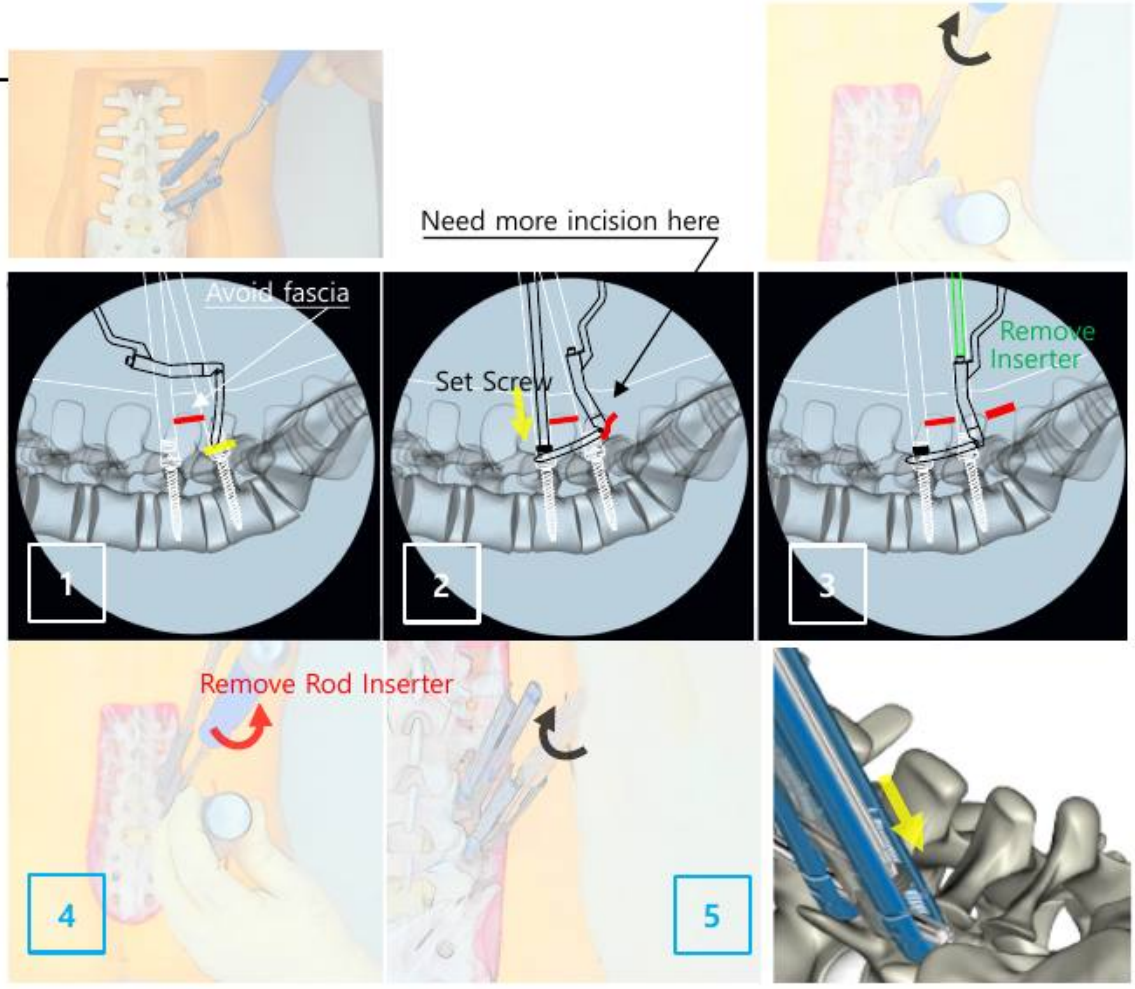


Rod Insertion and Reduction

After choosing the appropriate length rod, contour and corresponding rod holder, insert the connection end of the rod into the pocket of the rod holder ensuring that the notch on the connection end of the rod is facing up towards the handle of the rod holder.

Ensure rod is below the fascia. While rotating the rod, guide it toward the next screw channel. Visualize entry into the next screw. Do not release rod from Rod Holder until locking cap screws have been inserted and provisionally tightened. Fluoroscopy should be used to confirm the rod is positioned correctly and there is adequate overhang of the rod tip.

Attach locking caps to either Starter Driver option. Caps are held to the driver using a taper fit; hence, they are attached by a stab-to-grab action. Failure to adequately 'stab' lock ing cap can result in the caps falling off the driver inadvertently. Advance locking cap down the tower, ensuring the driver is colinear with the tower. Engage the threads of the tower with the locking cap. Continue to rotate clockwise until the cap secures the rod.

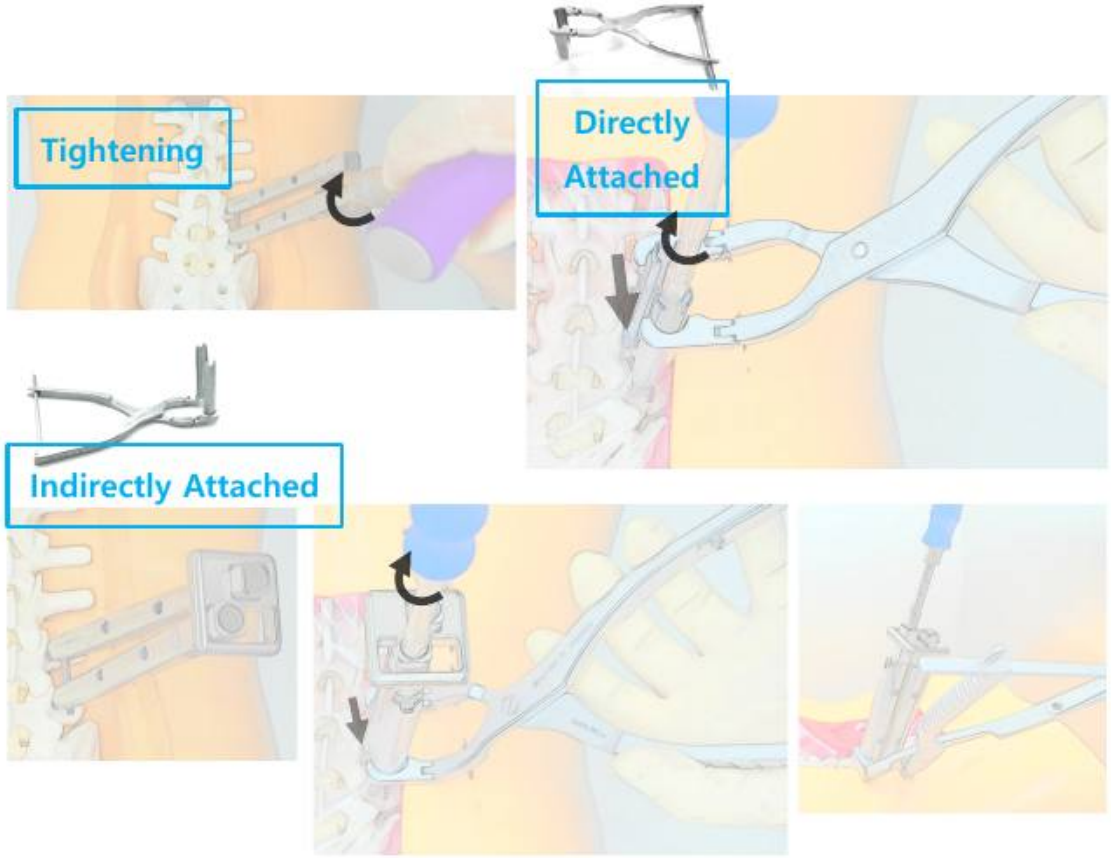


Compression

Slide the MIS sleeve over each screw tab to which you are going to apply compression forces. Application of compression forces without use of the sleeve in not recommended. The slot in the end of the sleeve is aligned and engages the rod. For compression, after all set screws have been preliminarily fixated to the rod, loosen the set screw of the pedicle screw to be adjusted using the set screw holder/driver. Slide the MIS pulcrum over the top of the sleeves and draw the sleeves together with either upward or downward force on the handle of the alignment tool.

Attach the compression instrument to the holes in the sleeves closest to the level of the skin. Compress against the appropriate modular body while tips remain on rod and re-tighten the set screw when desired compression has been achieved. Applying too much compression or distraction force to screws can result in pedicle fracture.

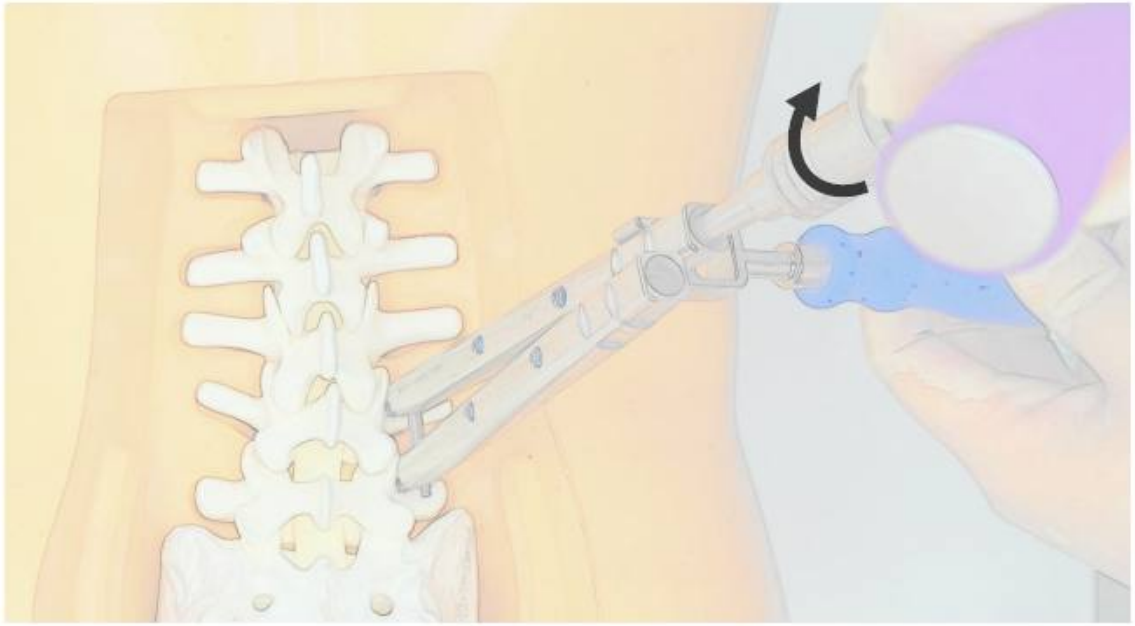
Compress the sleeves to the desired level and proceed to final tightening of Set Screws.



Final Tightening

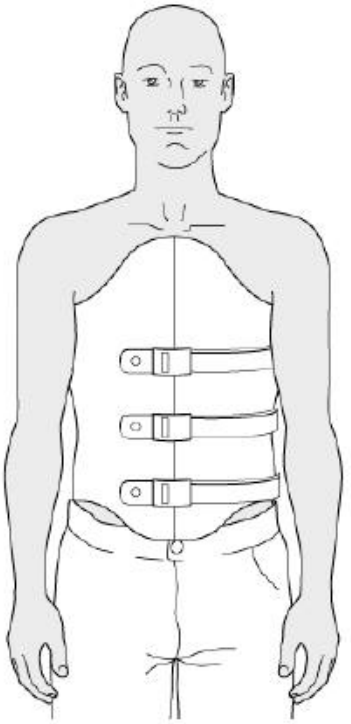
Position the counter torque wrench over the pedicle screw and rod making sure to engage tips to align rod within screw body. Place the set screw driver through the cannulation of the counter torque wrench and into the hex of the set screw. Turn the torque limiting handle clockwise and apply counter torque with the counter torque wrench to tighten the set screw to 100 in-lbs. The torque limiting handle will reach its maximum torque and release at 12 or 10.5 in-lbs.

Insert the set screw driver into the torque limiting handle by compressing connection mechanism. Ratcheting feature will not function properly unless driver is fully seated before releasing connection mechanism. Ensure the ratcheting dial is set to forward prior to engaging the set screw.



Postoperative care

After surgery patients typically exhibit all of the benefits of minimally invasive surgery. They are mobilized on the first postoperative day. A TLSO brace may be used at the surgeon's discretion.



1. Detailed instructions on the use and limitations of the implant should be given to the patient. The patient must be made aware of the limitations of the implant. Physical activity and load bearing have been implicated in premature loosening, bending, or fracture of internal fixation device.

2. Periodic X-rays for at least the first year postoperatively are recommended for close comparison with postoperative conditions to detect any evidence of changes in position, nonunion, loosening, and bending or cracking of components.

3. Surgical implants must never be reused. Any retrieved devices should never be reused in another surgical procedure. The retrieved parts should be handled and disposed of in such a manner as to ensure that reuse is not possible.

4. To allow the maximum potential for a successful surgical result, the patient or device should not be exposed to mechanical vibration that may loosen the device construct.

5. These implants are temporary internal fixation devices. Internal fixation devices are designed to assist in the stabilization of the operative site during normal healing process. After healing occurs, these devices serve no functional purpose and should be removed. In most cases removal is indicated because the implants are not intended to transfer or support forces developed during normal activities. If the device is not removed following completion of its intended use, complication may occur as follows:

a) Corrosion, with localized tissue reaction or pain. b) Migration of implant position resulting in injury. c) Risk of injury from postoperative trauma. d) Bending, loosening and/or breakage, which could make removal impractical or difficult. e) Pain, discomfort or abnormal sensations due to the presence of the device. f) Possible increased risk of infection. g) Bone loss caused by stress shielding.

Product Information

Implant

Item No.	Description
SA.605.5530	MIS Polyaxial Pedicle Screw Long Ø5.5mm x 30mm
SA.605.5535	MIS Polyaxial Pedicle Screw Long Ø5.5mm x 35mm
SA.605.5540	MIS Polyaxial Pedicle Screw Long Ø5.5mm x 40mm
SA.605.5545	MIS Polyaxial Pedicle Screw Long Ø5.5mm x 45mm
SA.605.5550	MIS Polyaxial Pedicle Screw Long Ø5.5mm x 50mm
SA.605.6535	MIS Polyaxial Pedicle Screw Long Ø6.5mm x 35mm
SA.605.6540	MIS Polyaxial Pedicle Screw Long Ø6.5mm x 40mm
SA.605.6545	MIS Polyaxial Pedicle Screw Long Ø6.5mm x 45mm
SA.605.6550	MIS Polyaxial Pedicle Screw Long Ø6.5mm x 50mm
SA.605.6555	MIS Polyaxial Pedicle Screw Long Ø6.5mm x 55mm
SA.605.7535	MIS Polyaxial Pedicle Screw Long Ø7.5mm x 35mm
SA.605.7540	MIS Polyaxial Pedicle Screw Long Ø7.5mm x 40mm
SA.605.7545	MIS Polyaxial Pedicle Screw Long Ø7.5mm x 45mm
SA.605.7550	MIS Polyaxial Pedicle Screw Long Ø7.5mm x 50mm
SA.605.7555	MIS Polyaxial Pedicle Screw Long Ø7.5mm x 55mm
SA.M55.0040	MIS ROD 5.5 40mm
SA.M55.0045	MIS ROD 5.5 45mm
SA.M55.0050	MIS ROD 5.5 50mm
SA.M55.0055	MIS ROD 5.5 55mm
SA.M55.0060	MIS ROD 5.5 60mm
SA.M55.0065	MIS ROD 5.5 65mm
SA.M55.0070	MIS ROD 5.5 70mm
SA.M55.0075	MIS ROD 5.5 75mm
SA.M55.0080	MIS ROD 5.5 80mm
SA.M55.0085	MIS ROD 5.5 85mm
SA.M55.0090	MIS ROD 5.5 90mm
SA.M55.0100	MIS ROD 5.5 100mm
SA.M55.0110	MIS ROD 5.5 110mm
SA.M55.0120	MIS ROD 5.5 120mm
SA.M55.0130	MIS ROD 5.5 130mm
SA.M55.0140	MIS ROD 5.5 140mm
SA.M55.0150	MIS ROD 5.5 150mm
SA.M55.0300	MIS ROD 5.5 300mm
10.22.020	Set Screw ⌀10x3.8mm





Product Information *continued*

Instrument

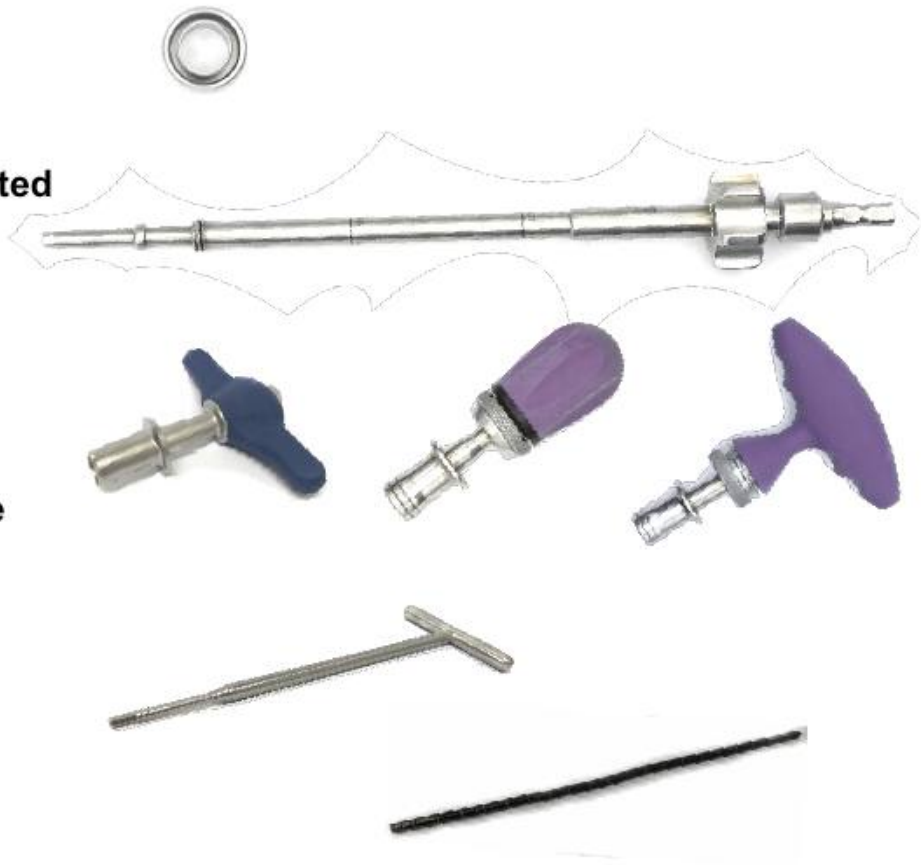
Item No.	Description
	Needle
	K-wire
20505600	Multi Level Sleeve
SI.CP6.0001	Modular Cannulated Tap 5.5
SI.CP6.0002	Modular Cannulated Tap 6.5
SI.CP6.0003	Modular Cannulated Tap 7.5



Product Information *continued*

Instrument *Continued*

Item No.	Description
SI.MP6.0009	MIS Cap
SI.MP6.0004	MIS Polyaxial Screw Driver Cannulated
91200100	Cannulated T-Handle
G118980	Cannulated T-Ratchet Handle
G176916	Extended Palm Style Ratchet Handle
SI.AP6.0003	Anchor Adjuster
23001001	20cm Template rod



Product Information *continued*

Instrument *Continued*

Item No.	Description
20505403	Rod holder(Long)
20507600	Driver for Rod Inserter(type 1)
SI.AP6.0025	Set Screw Driver I-Type
SI.AP6.0026	Set Screw Driver T-Type
SI.MP6.0005	MIS Sleeve
SI.MP6.0006	MIS Fulcrum



Product Information *continued*

Instrument *Continued*

Item No.	Description
20506500	Compressor New
SI.MP6.0007	MIS Anti Torque Handle
SI.AP6.0024	Modular Set screw
SI.ZH1.0003	Torque Limiter Balanced silicone T-handle(10.5Nm)
SI.MP6.0008	MIS Guide Breaker
B211-10100	French Bender
20506600	Distractor New





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